

ASSIGNMENT 3 - IMPLEMENTATION

NYCTLC TAXI FARE

PREDICTION

From Proposal to Production: Implementing GBDT for Dynamic Pricing & Fleet Optimisation

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Assessment Declaration:

I declare that in submitting all work for this assessment I have read, understood and agree to the content and expectations of the Assessment declaration.

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1. Introduction

1.1 The Business Problems

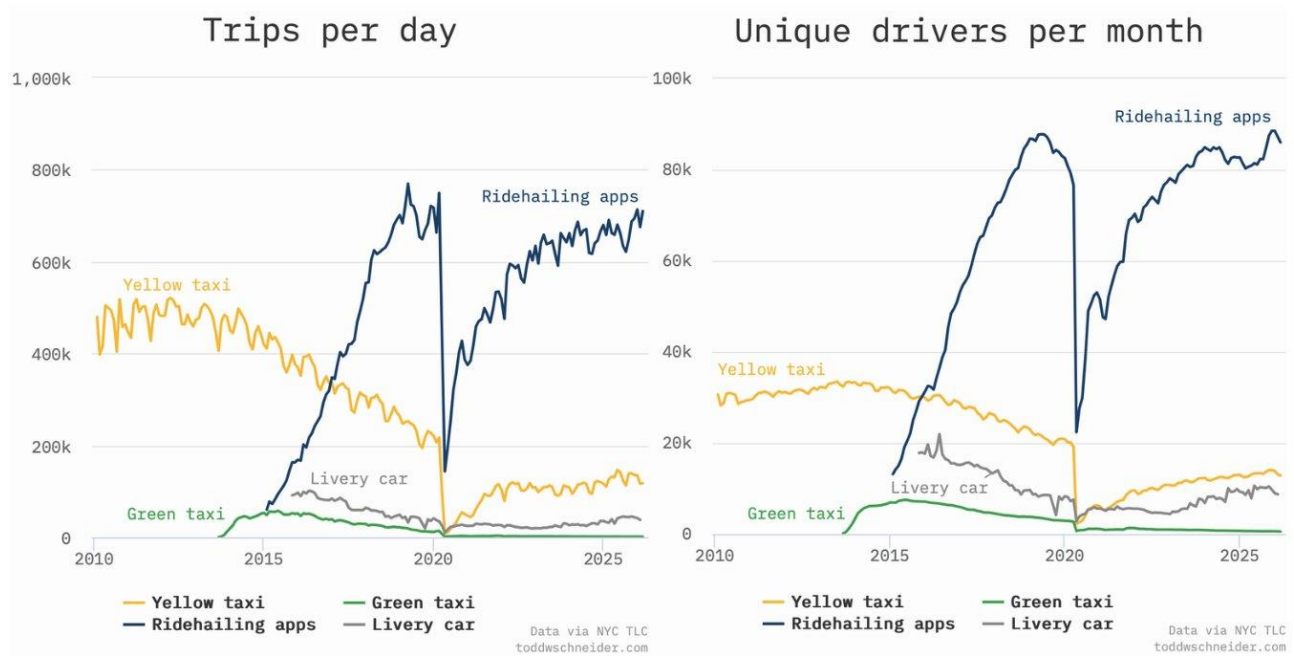
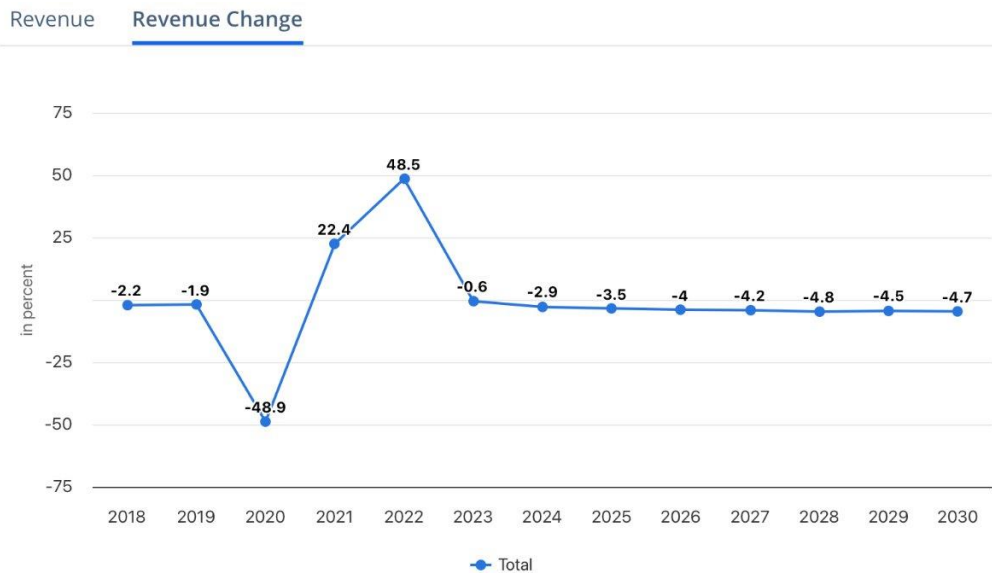


Figure 1: Trips per day & Unique driver per month (Yellow Taxi, Green Taxi, Ride-hailing App, Livery Car) (adapted from Toddwschneider 2026).



Most recent update: thg 6 2025
Source: Statista Market Insights

Figure 2: Revenue Change of Taxi in United States (2018-2030) (adapted from Statista 2025).

The New York City Taxi & Limousine Commission (NYCTLC) regulates more than 13,000 licensed yellow-cab medallions that once defined urban mobility in the city. Over the past decade, this fleet has lost substantial market share to app-based ride-hailing competitors such as Uber and Lyft (*Figure 1*; NYCTLC 2026). The decline is structural rather than cyclical: industry projections estimate a 4.7% annual revenue contraction through 2030 (*Figure 2*), while consumer research indicates that 83% of riders now prioritise convenience over price (McKinsey&Company 2017). Three interlocking operational failures drive this erosion.

1. **Fare prediction inefficiency.** The legacy meter applies flat, fixed per-mile pricing that cannot respond to real-time demand. During peak periods the fleet forfeits revenue it could legitimately capture, while riders face no transparent expectation of trip cost before boarding.
2. **Inefficient taxi allocation.** Drivers rely on habit and intuition to locate passengers, circling high-traffic corridors with empty vehicles. This produces idle rates exceeding 30%, wasting fuel, driver income, and vehicle availability.
3. **Demand-supply mismatch.** Without a systematic peak/off-peak strategy, supply is poorly matched to demand across time and space. Customers experience long wait times and defect to ride-hailing apps that resolve matching algorithmically within minutes.

1.2 Motivation for Machine Learning

These problems share a common root: pricing and dispatch decisions are made without quantitative foresight. Machine learning directly addresses this gap by converting historical trip records into a predictive engine that estimates fares before a trip begins (Alam et al. 2025). The benefits are threefold. A reliable fare model enables dynamic, defensible pricing (Mehta et al. 2022); an interpretable feature ranking reveals which factors genuinely drive cost, informing dispatch (Ma et al. 2026); and an accurate baseline lets management apply conservative surge multipliers without overcharging customers or underpaying drivers (Mehta et al. 2022; Alam et al. 2025). Crucially, the model produces decisions that are auditable, satisfying the strict public transparency mandates of NYC Local Law 35 (2022) and aligning with the rigorous centralized evaluation standards enforced by the city's newly established Office of Algorithmic Accountability (Gutiérrez 2025).

1.3 Chosen Model Type

The task is a **supervised regression problem**: predict a continuous target with fare amount in US dollars from trip attributes. The report selected a Gradient Boosted Decision Tree (GBDT),

implemented through Microsoft Azure Machine Learning's Boosted Decision Tree Regression module. Regression suits the problem because the business needs an exact dollar estimate rather than a category, and GBDT specifically suits NYC fare behaviour because that behaviour is sharply non-linear, as the following sections detail.

2. Model Design and Implementation

2.1 Model Architecture and Suitability

A GBDT builds an ensemble of shallow decision trees in sequence. Each successive tree is trained on the residual errors of the trees before it, so the model progressively corrects its own mistakes (Hong et al. 2021; Chen et al. 2024). This sequential error-correction is the decisive property for NYC fares. A two-kilometre trip through Manhattan at 5 PM during rush hour costs materially more, and takes far longer, than the same two kilometres at 3 AM. Standard linear regression cannot capture this interaction without extensive manual feature engineering, and Random Forests - which average independent trees - smooth over the sharp demand spikes that characterise peaked hours or airport traffic. GBDT, by contrast, models these non-linear spikes natively, which is why peer-reviewed work (Ke et al. 2017; Si et al. 2017) favours it for high-variance tabular prediction.

The model ingests 10 engineered features spanning three categories: spatial (Borough, Zone, PULocationID), temporal (Month, DayOfWeek, HourOfDay), trip-intrinsic (trip_distance, trip_duration_minutes, RatecodeID), and numbers of passengers (passenger_count). It outputs a predicted fare in dollars, a feature-importance ranking, and an implied confidence level suitable for pricing decisions.

2.2 Integration into the Existing Business

The model is designed to slot into NYCTLC's operational workflow rather than sit beside it. A dispatcher-facing command centre consumes the model's zone-level demand predictions to recommend fleet repositioning, while a driver-facing mobile application surfaces per-trip fare predictions and earnings forecasts (*Figure 8&9*). Both layers call the same trained model, ensuring that the macro-level dispatch strategy and the micro-level driver decisions are derived from one consistent source of truth. Because Azure ML exposes the trained pipeline as a managed REST endpoint, integration with these front-end applications requires only standard API calls, with Azure handling provisioning, scaling, and monitoring (Moutaouakal and Baïna 2023).

2.3 Implementation

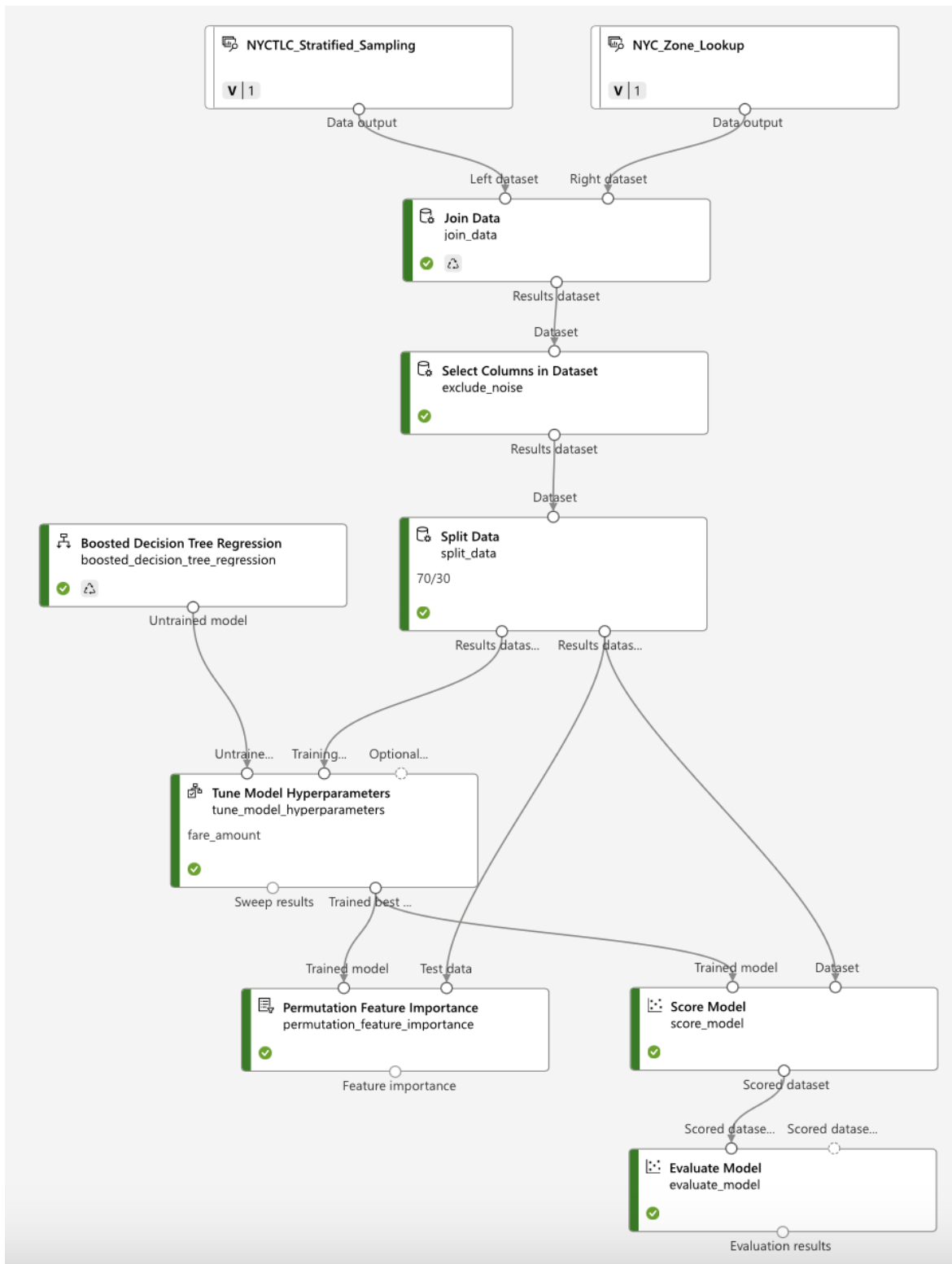


Figure 3: NYCTLC's Azure Machine Learning Pipeline.

Implementation followed a seven-stage pipeline built end-to-end in Azure Machine Learning Designer:

- **Pre-processing (Tableau Prep - Appendix 1):** Twelve monthly 2019 trip files were unioned and cleaned across roughly twenty steps, including a Tukey IQR capping method that addressed the 10–15% of null and extreme values while retaining 90% of data integrity (from ~90M to ~84M rows) (Pavelchek et al. 2023).
- **Stratified sampling (Python & Jupyter - Appendix 2):** The 12.42 GB, 84-million-row dataset exceeded local memory, so a streaming chunked stratified sample of roughly 5.05 million rows was drawn across all twelve months and 242 of 263 zones, preserving representativeness.
- **Join & feature selection (Azure ML - Figure 3):** Pickup location IDs were joined to the TLC Zone Lookup to add borough and zone names; leakage-prone and noise columns (`tpep_pickup_datetime`, `total_amount`, `tip_amount`, `service_zone`, `payment_type`) were dropped, yielding 10 clean features and roughly 60× faster training (reduced from +8hours to under 20 minutes).
- **Split Data (Figure 3):** Data was split 70/30 (70% training dataset vs. 30% testing dataset).
- **Tune Model Hyperparameters (Figure 3):** The Boosted Decision Tree was trained and optimised via the Tune Model Hyperparameters module using a random sweep over 20 runs against RMSE. The tune helps identify the optimal configuration to improve +7% accuracy and Uber, Lyft, DoorDash also utilize tune hyperparameters before deploying (Xia et al. 2017).
- **Permutation Feature Importance (Figure 3):** The permutation validates which features genuinely drive predictions by measuring the increase in error when each feature's values are randomly shuffled on the held-out test set (Bhuiyan et al. 2026).
- **Score & Evaluate (Figure 3):** The tuned model was scored on the hold-out 30% test set, evaluated on standard regression metrics, and interrogated with Permutation Feature Importance to validate which features genuinely drive predictions.

3. Model Results and Analysis

3.1 Performance Results

This model was built to answer one question: can a machine really predict New York City taxi fare accurately enough to change how the business operates? We put that to test by running 1.5M completely unseen trip records through GBDT model. Results from the pipeline showed exceptional predictive records as shown in Table 1 below.

Metric	Value	Interpretation
Mean Absolute Error (MAE)	\$0.54	Average prediction is within 54 cents of the true fare
Root Mean Squared Error (RMSE)	2.74	Slightly above MAE, signaling a small number of outliers
R ² (Coefficient of Determination)	0.947	Model explains 94.7% of fare variance
Relative Squared Error	0.053	5.3% normalised error - Very Low
Relative Absolute Error	0.071	7.1% relative deviation from the mean

Table 1: The Result of NYCTLTC's Evaluation Model Performance (adapted from *Appendix 3*).

A feature importance analysis as conducted to better understand the model's behaviour. Results show trip distance and trip duration are the primary drivers, making up for 85% of the model's predictions. Distance contributes approximately 46%, while duration contributes around 39%. Rate code followed at 1.47, with zone, borough, and location identifiers contributing to the remaining spatial signal, and temporal features (hour, month, day-of-week) contributing marginally.

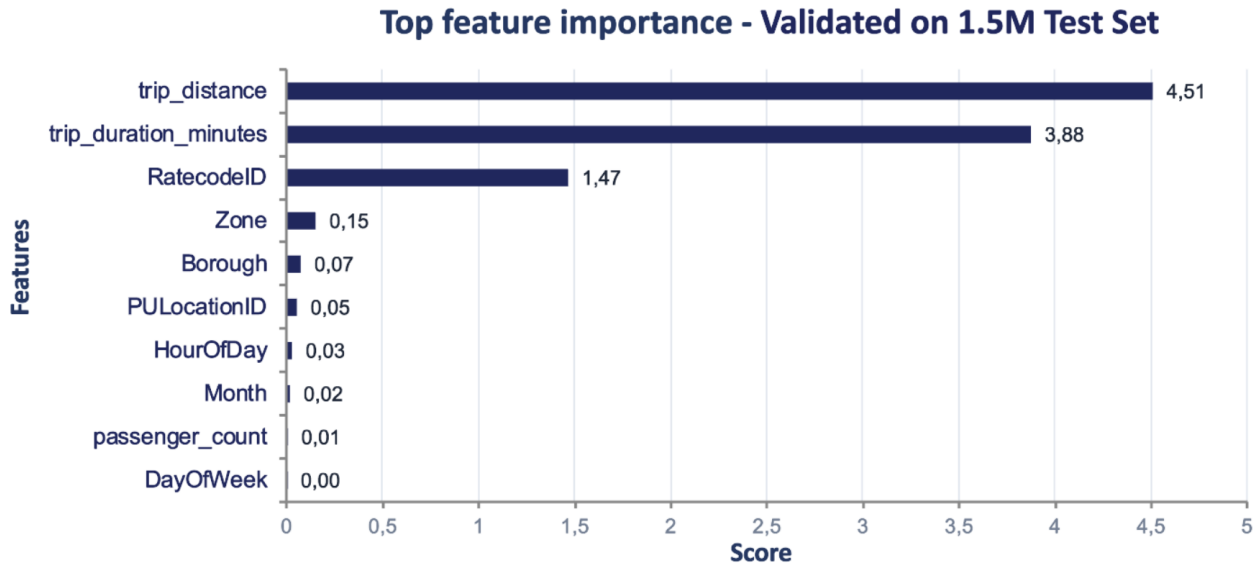


Figure 4: The Result of NYCTLC’s Top Feature Importance (adapted from *Appendix 4*).

The Score Model output reveals a right-skewed fare distribution: a mean of \$13.29 against a median of \$9.43, with the majority of trips clustering between \$7 and \$27 and a long tail extending to a \$190.52 maximum (*Appendix 5*). This shape is exactly what the Permutation Feature Importance ranking predicts (*Figure 4*). Because distance and duration jointly drive 85% of the prediction, the bulk of short, low-mileage Manhattan trips collapses into the dense \$7–\$27 cluster, while the long tail corresponds to the high-distance and airport trips that the third-ranked feature, RatecodeID, isolates via flat rates such as the \$52 JFK fare. In other words, the scored output distribution and the feature ranking tell the same story from two angles: the model's predictions are concentrated where the dominant features (distance, duration) are small, and dispersed precisely where the rate-code feature flags premium trips. The negative minimum of −\$13.38 reflects refund records in the source data rather than model failure, and is clipped at the regulated base fare in production.

3.2 Alignment with the Business Problem

The model learnings map cleanly onto each of the three business problems, and the feature analysis explains why.

Top 2 Features Drive 85% of Fare Prediction

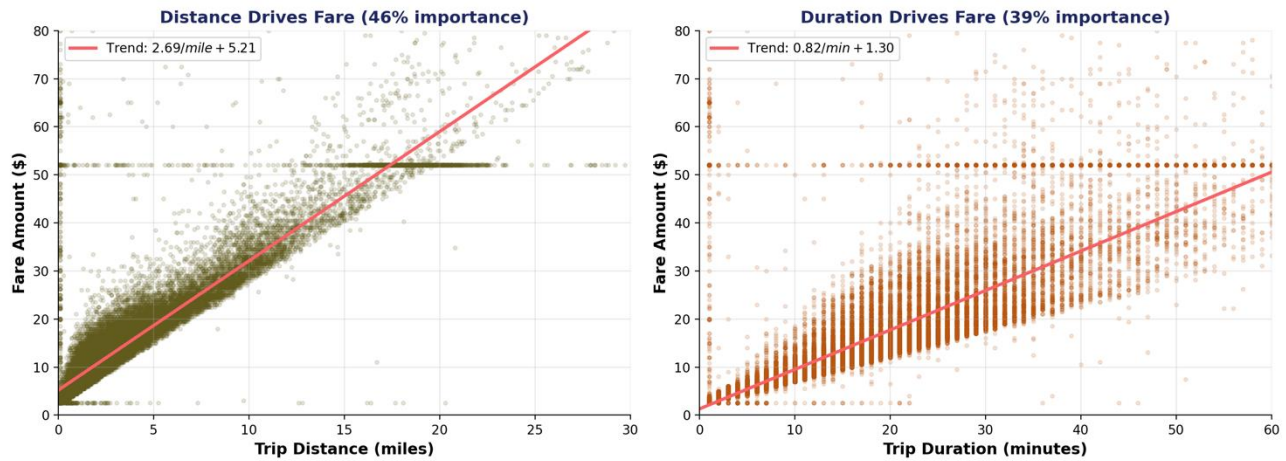


Figure 5: Scatter Analysis of Fare Amount vs. Trip Distance & Trip Duration.

Fare inefficiency is resolved directly. It reveals the same patterns for both variables: the more the further the distance, the more customers pay with a base rate of \$5.21 and an additional \$2.69 per mile. Similarly, the longer the trip duration, the more they pay with 0.89 cents per minute on top of \$1.30 base rate (Figure 5). With 94.7% variance (Table 1), it enables the controlled use of dynamic pricing to maintain operational efficiency while remaining fair to customers, for example a moderate 1.2x adjustment during peak hours without the risk of over- or under-pricing.

Airport Trips Command 3-4× Higher Fares Than Standard Trips

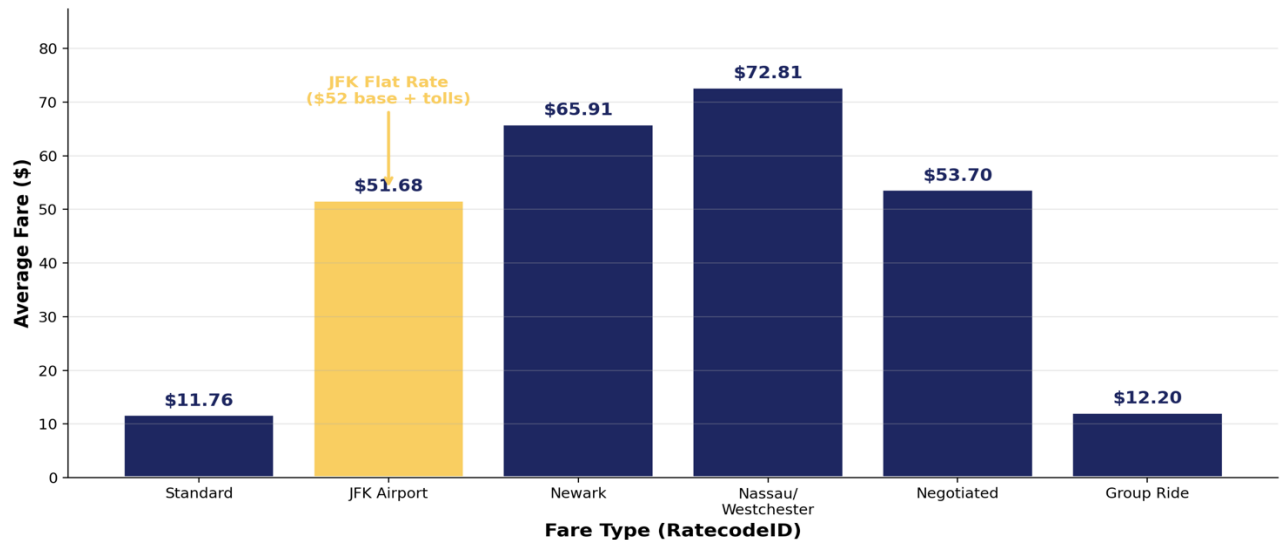


Figure 6: Bar Chart of RatecodeID Determined Premium-Typed Trip by Average Fare Amount.

Taxi allocation is resolved through the rate-code and spatial signal. Airport trips, in particular, carry a significant premium. A typical trip in the city (Ratecode 1) averages around \$11.76, while

trips to JFK Airport (Ratecode 2), Newark (Ratecode 3) and Nassau/Westchester (Ratecode 4) are 4.4x, 5.6x and 6.2x higher consecutively (*Figure 6*). Dispatchers can therefore direct vehicles toward high-value corridors by real-time revenue heatmap (*Appendix 6*) rather than letting drivers guess, reducing idle time by an estimated 15–20%.

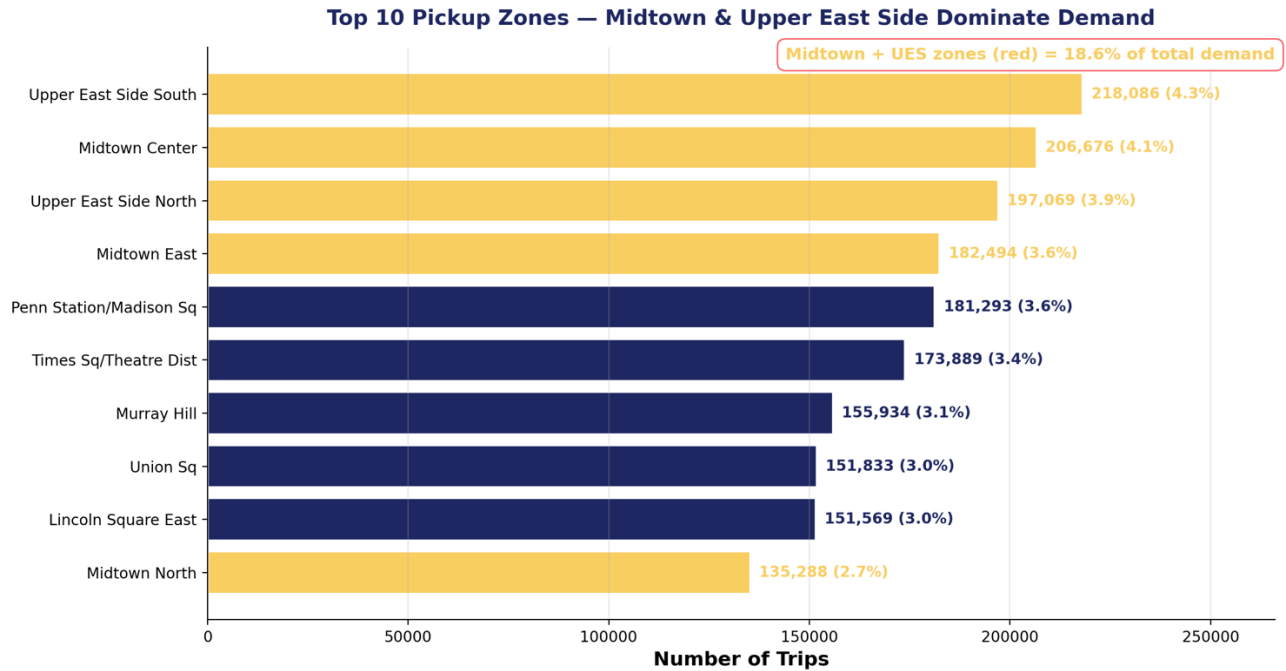


Figure 7: Top 10 Pickup Zones in NYC.

Demand-supply mismatch is resolved by combining spatial and temporal features. Four core Manhattan zones generate 15.9% of all demand within a one-kilometre radius (*Appendix 7; Figure 7*). Coupling this spatial concentration with the hour-of-day signal lets the fleet pre-position before peaks occur (*Appendix 8 & 9*), shifting from passive hailing to proactive matching and targeting a 2–3-minute wait time competitive with ride-hailing apps.

In conclusion, fleet operations can move from reactive decision-making toward a more proactive strategy, instead of relying on drivers to wait for street hails, operators can use predictions to position vehicles in high-demand areas ahead of peak periods, including key urban corridors and airport routes. This approach supports more efficient fleet utilization and improved service availability. It also enables the controlled use of dynamic pricing to maintain operational efficiency while remaining fair to customers.

3.3 Challenges and Resolutions

Four substantive challenges arose during implementation. The raw datetime column generated roughly 199,000 spurious one-hot features and slowed training dramatically; dropping the raw timestamp in favour of engineered month/day/hour fields restored speed. A subtler risk was data leakage: total_amount and tip_amount exist only after a trip concludes, so using them to predict fare would constitute training on the answer. Both were removed, alongside two further redundant columns, leaving ten clean features. Azure ML Designer could not efficiently process the full 84-million-record dataset (12.42 GB scale). Therefore, a Python-based stratified sampling approach was used to create a representative 5-million-record subset, reducing computational load while preserving key spatial and temporal patterns. Finally, the TLC Zone Lookup table was joined to enrich location IDs with borough and zone information, addressing both the lack of geographic meaning and the strong regional imbalance in trip distribution. Collectively these steps cut training time from over eight hours to under twenty minutes.

3.4 Integrated Azure Model into NYCTLC systems

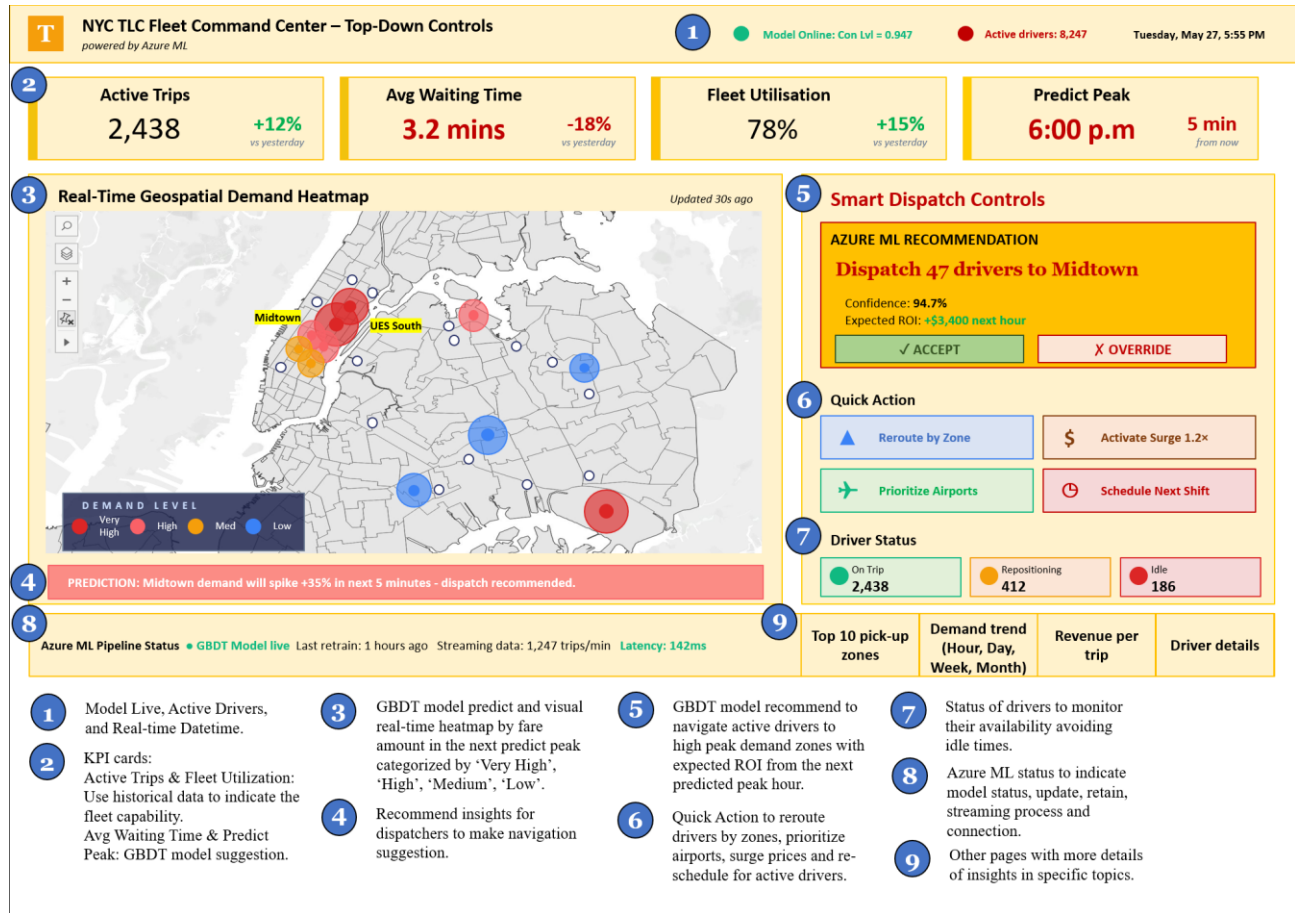


Figure 8: Fleet Command Center Monitoring Dashboard for NYCTLC's Dispatcher.

The trained GBDT model is operationalised through two complementary interfaces that convert its predictions into front-line decisions. The Fleet Command Center serves dispatchers with a top-down view: a **real-time geospatial heatmap** renders predicted fare-demand by zone ('Very High' to 'Low'), while four **KPI cards** track active trips, fleet utilisation, average wait time, and the next predicted peak (*Figure 8*). The model's core value surfaces in the **Smart Dispatch panel**, which translates raw predictions into a concrete recommendation "Dispatch 47 drivers to Midtown, confidence 94.7%, expected ROI +\$3,400" that a dispatcher accepts or overrides in one click (*Figure 8*). **Quick Actions** and a **live driver-status breakdown** close the loop between prediction and action (*Figure 8*).

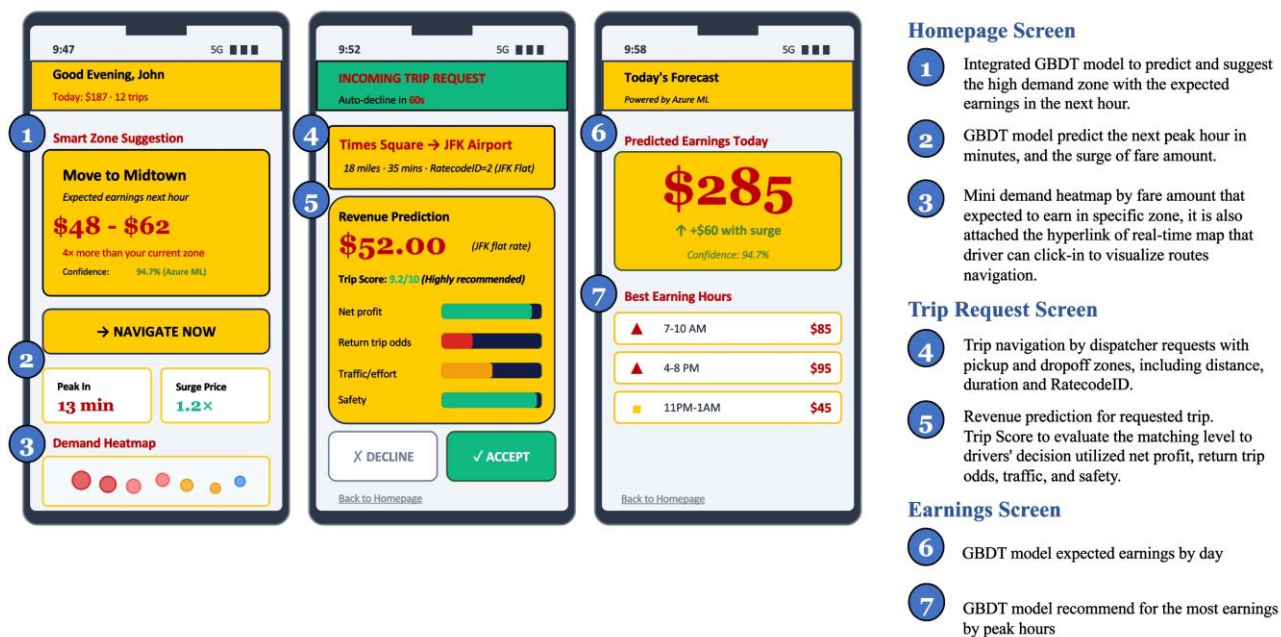


Figure 9: Driver App for NYCTLC’s Yellow Cap Taxi Drivers.

The Driver App mirrors this intelligence bottom-up, serving the individual driver who executes those recommendations. It surfaces a **Smart Zone Suggestion** with expected hourly earnings, a per-trip Revenue Prediction, and a Trip Score that critically blends the fare prediction with net profit, return-trip odds, traffic effort, and safety, acknowledging that a high gross fare is not always the best trip (Figure 9). An earnings forecast and best-earning-hours ranking help drivers plan their shift around predicted peaks (Figure 9).

Together the two systems form a closed feedback loop: the same model that advises macro-level fleet allocation also guides micro-level driver choices, ensuring strategy and execution derive from one consistent source of truth. This dual-layer integration demonstrates that the model is not a static artefact but a live decision engine embedded in NYCTLC's daily operations.

4. Ethical Considerations and Reflective Insight

4.1 Ethical Implications and Potential Biases

The most serious ethical risk is an algorithmic feedback loop. The 2019 training data is heavily concentrated on trips in Manhattan's areas, roughly 95% of trips. If the model is allowed to optimise for profit unchecked, it will route more drivers into these high-margin zones, creating less service coverage in the Bronx and outer boroughs. This reinforces existing inequalities in access to transportation (Pandey and Caliskan 2020), as the model would continuously learn data collected from these high-demand areas like Manhattan, the bias would compound over successive retraining cycles (Jonker and Julie 2026). This means the problem is not just theoretical; it is a real risk when using demand-based optimisation on data that is uneven.

4.2 Accountability and Transparency Measures

Two design choices make the system accountable. First, the deliberate selection of GBDT over a black-box neural network means feature-importance scores are fully auditable (Fang et al. 2020), where the compliance team can review the model and detect any potential bias across different boroughs (Hong et al. 2021; Chen et al. 2024). Second, the system includes fairness-based incentives. When service coverage in outer boroughs drops below a threshold, the model automatically applies fare adjustments and driver incentives to encourage better service distribution and reduce historical bias of imbalance. On privacy, the model uses only anonymous trip records with no personally identifiable information, and a public fare-prediction dashboard is planned, so riders can see why a given surge applies. This aligns with responsible AI guidance from UNESCO, which stress explainability, human oversight, traceability, and ongoing risk management across the system (UNESCO n.d).

4.3 Limitations and Opportunities for Improvement

There are three limitations, the model predicts fare, not trip count, fare is used as a proxy for demand, which captures economic value but does not directly measure trip volume, hence a dedicated demand-forecasting model is the natural next step. Secondly, external factors such as weather, events, and traffic data were not included, which can significantly impact demand capacity by $\pm 30\%$. Lastly, the model is trained on pre-COVID 2019 data, which may further limit the model's applicability on current travel behavior. A minor technical issue is that the predicted minimum is negative ($-\$13.38$), mainly due to refund records in the source data rather than model failure. This can be handled by

setting a minimum fare threshold based on the regulated base fare. Acknowledging these limits openly builds trust and produces a clearer investment roadmap.

5. Competitive Advantage and Future Opportunities

5.1 Competitive Advantage

The solution gives NYCTLC pricing and dispatch parity with Uber and Lyft while preserving two structural advantages those platforms cannot match. The comparison in *Table 2* below summarises the positioning.

Capability	Uber / Lyft	NYCTLC (Azure Solution)
Fleet Size in NYC	18,000+	13,000 Medals
Dynamic pricing	Yes	Yes
Data-driven dispatch	Yes	Yes
Regulated brand trust	Mixed	High
Commission cost to driver	25–30%	0% (lower net fare)

Table 2: Comparative Advantages of NYCTLC vs. Uber/Lyft.

Because medallion taxis charge no platform commission, NYCTLC can offer a lower net fare to riders or higher take-home pay to drivers at parity pricing (*Table 2*). Combined with regulated brand trust and now algorithmic dispatch, this converts a regulatory constraint into a differentiator. The projected financials are compelling, a \$0.45M first-year investment against \$2.1M in annual benefits yields roughly 366% first-year ROI (*Figure 10*).

Estimated annual financials for the Azure ML solution

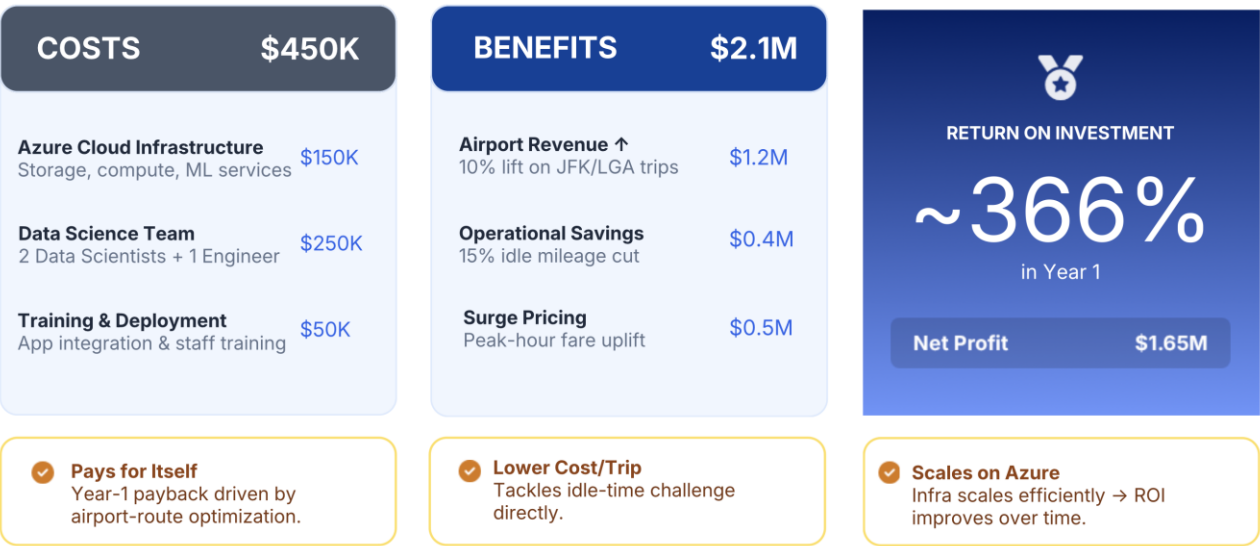


Figure 10: Cost Analysis in Year 1 of Azure AI Model Application.

5.2 Future Opportunities

The roadmap extends the model along three horizons. In months 6–12, Azure Stream Analytics enables real-time live driver heatmaps and sub-minute forecasting (*Figure 8 – applied for Node 3*). In year two, integrating weather APIs, an events calendar, and traffic feeds is expected to lift accuracy by capturing the $\pm 30\%$ external swing the current model misses. From year three and beyond, the platform can support EV route optimisation aligned with the city's sustainability goals and multi-modal integration with subway and bus networks, positioning NYC TLC not merely as a taxi operator but as a data-driven mobility platform. The roadmap extends the platform’s two structural advantages regulated trust and zero commission across three horizons, indicated in details in *Appendix 10*.

6. Conclusion

6.1 Summary of Key Findings

Gradient Minds delivered a fully functional GBDT fare-prediction model on Azure Machine Learning, trained on a representative 5-million-row sample and validated on 1.5 million held-out trips. The model achieves an R^2 of 0.947 and a mean absolute error of \$0.54, with permutation analysis confirming that distance and duration drive 85% of predictions and that rate code captures the airport premium. These results map directly onto the three business problems, enabling dynamic pricing, data-driven allocation, and proactive demand-supply matching.

6.2 Recommendations

- **Deploy as a managed Azure endpoint** and run a controlled pilot with 100 drivers before fleet-wide rollout, validating the 1.2× surge and dispatch logic via A/B testing (Microsoft n.d).
- **Embed the fairness caps and monthly retraining** from day one, so equity and concept-drift safeguards are operational rather than aspirational (Uddin et al. 2024).
- **Fund the Phase 2 data partnerships** (weather, events, traffic) to close the largest accuracy gap, and commission a dedicated demand-count model to complement the fare predictor (Yu et al. 2021; Tan and Tao 2023).

In conclusion, the model is technically sound, ethically responsible, and offers a clear competitive advantage. It provides NYCTLC with a reliable and transparent approach to compete with ride-hailing platforms, while still meeting its responsibility to serve the public.

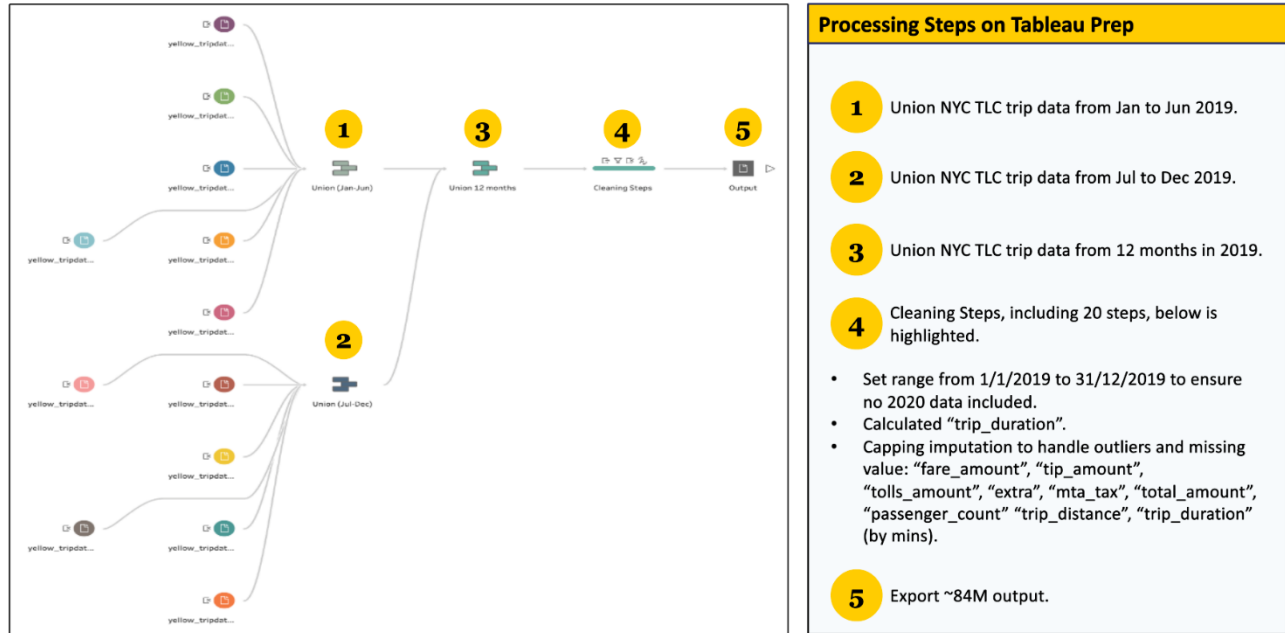
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8. Appendices



Appendix 1: Tableau Prep's Functional Data Workflow (NYCTLC's data from Jan-Dec 2019).

```
CHUNK_SIZE = 500_000 #each month sampled by 500K
SAMPLE_FRACTION = 0.06 # ~6% = 5M rows for file 84M
OUTPUT_FILE = "NYCTLC_Stratified_Sample.csv"

# Keep add-valued column (Using Capped versions)
COLUMNS_TO_KEEP = [
    'tpep_pickup_datetime',
    'PULocationID',
    'passenger_count (Capped)',
    'trip_distance (Capped)',
    'fare_amount (Capped)',
    'tip_amount (Capped)',
    'total_amount (Capped)',
    'trip_duration_minutes (Capped)',
    'payment_type',
    'RatecodeID'
]

def parse_vietnamese_datetime(series):
    # Replace SA -> AM, CH -> PM
    series = series.astype(str).str.replace(' SA', ' AM').str.replace(' CH', ' PM')
    # Parse by format DD/MM/YYYY
    return pd.to_datetime(series, format='%d/%m/%Y %I:%M:%S %p', errors='coerce')

if Path(OUTPUT_FILE).exists():
    Path(OUTPUT_FILE).unlink()
    print("Removed old output file")

sampled_chunks = []
total_processed = 0
total_sampled = 0
header_written = False
error_chunks = 0

print("Starting chunked stratified sampling...")
print("="*60)
```

1 Set Chunk Size = 500,000

2 Keep Capped Variables

3 Handle datetime

4 Remove existing old output

5 Process Sampled Chunks

```
#Read Errors
reader = pd.read_csv(
    "NYCTLC.csv",
    chunksize=CHUNK_SIZE,
    usecols=COLUMNS_TO_KEEP,
    on_bad_lines='skip', # Skip bad lines instead of stop
    encoding='utf-8',
    encoding_errors='ignore', # Ignore encoding errors
    low_memory=False
)

# Create 'Month'/'HourOfDay'/'DayOfWeek' for Stratified Sampling
for i, chunk in enumerate(reader):
    pickup_dt = parse_vietnamese_datetime(chunk['tpep_pickup_datetime'])
    chunk['Month'] = pickup_dt.dt.month
    chunk['HourOfDay'] = pickup_dt.dt.hour
    chunk['DayOfWeek'] = pickup_dt.dt.dayofweek

    # Drop rows datetime errors
    chunk = chunk.dropna(subset=['Month'])
    if len(chunk) == 0:
        error_chunks += 1
        continue

    #Stratified Sampling by Month + PULocationID
    sampled = chunk.groupby(['Month', 'PULocationID'], group_keys=False).apply(
        lambda x: x.sample(frac=SAMPLE_FRACTION, random_state=42) if len(x) > 0 else x
    )

    sampled.to_csv(OUTPUT_FILE, mode='a', headernot header_written, index=False)

    header_written = True
    total_processed += len(chunk)
    total_sampled += len(sampled)

    if (i + 1) % 10 == 0:
        print(f"Chunk {i+1}: processed {total_processed:,} | sampled {total_sampled:,}")

print("\n" + "="*60)
print(f"Total processed: {total_processed:,}")
print(f"Total sampled: {total_sampled:,}")
print(f"Output: {OUTPUT_FILE}")
print(f"File size: (Path(OUTPUT_FILE).stat().st_size / (1024*3)).2f GB")

✓ 3m 53s
```

6 Read errors not to stop reading since meeting 1 row error

7 Create 'Month' 'HourOfDay', 'DayOfWeek'

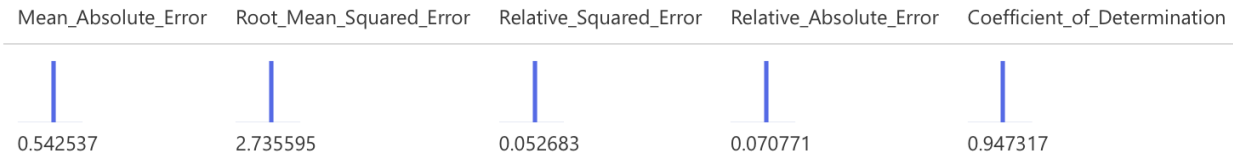
8 Drop error datetime

9 Run Stratified Sampling by month and zone

Appendix 2: Stratified Sampling using Python & Jupyter Notebook.

Rows ⑦
1

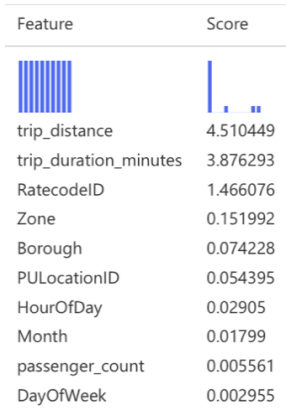
Columns ⑦
5



Appendix 3: Outcome Performance of NYCTLC’s Evaluation Model in Azure AI.

Rows ⑦
10

Columns ⑦
2



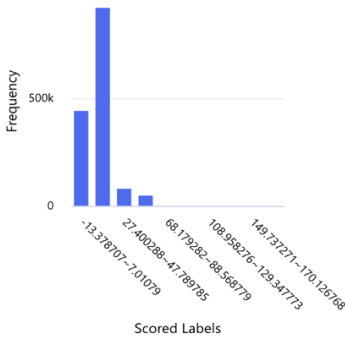
Appendix 4: Outcome of NYCTLC’s Pertumation Feature Importance in Azure AI.

Scored Labels

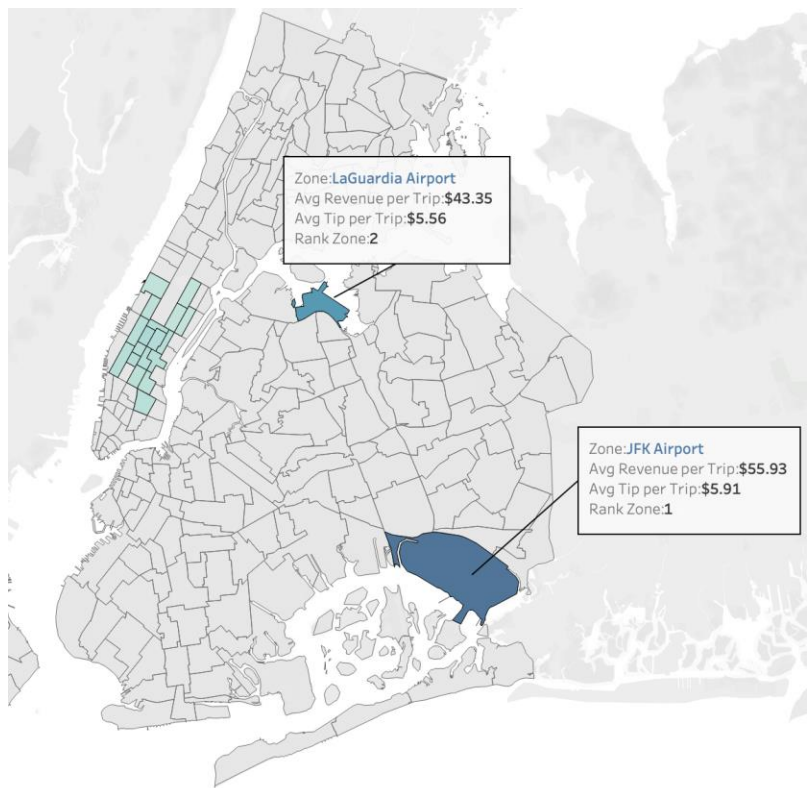
Statistics

Mean	13.2928
Median	9.4349
Min	-13.3787
Max	190.5163
Standard deviation	11.6139
Unique values	1276638
Missing values	0
Feature type	Numeric Score

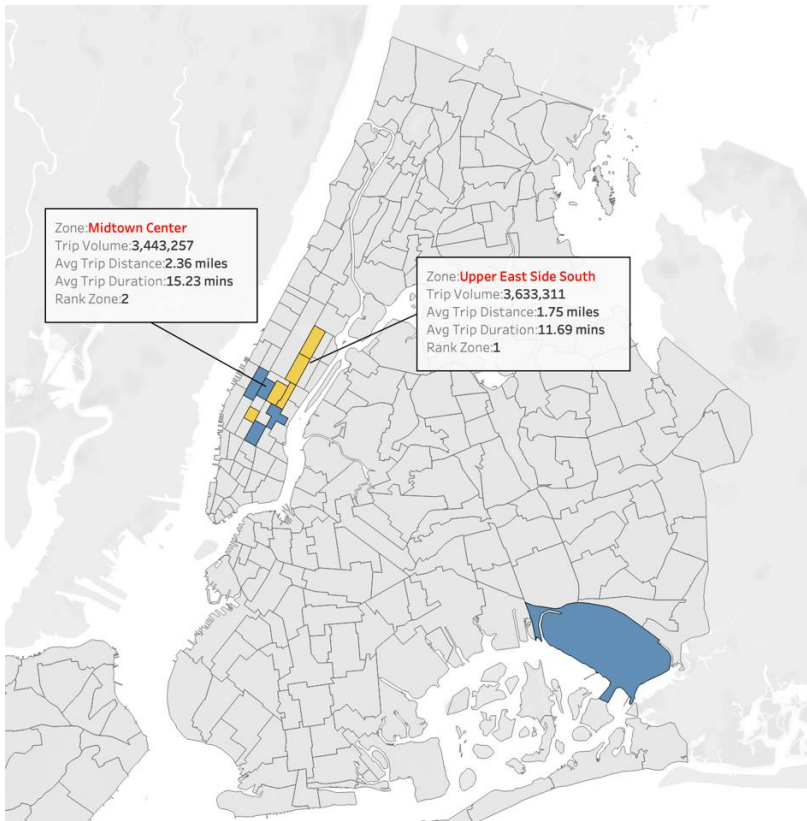
Visualizations



Appendix 5: Outcome of NYCTLC’s Score Model in Azure AI.

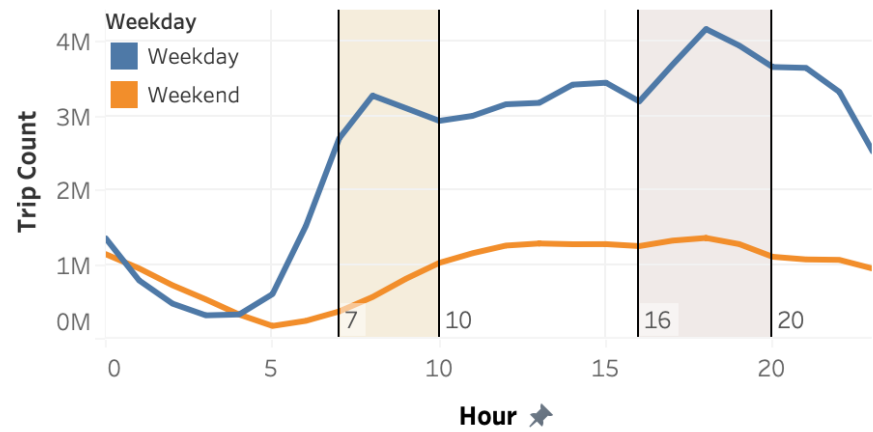


Appendix 6: Real-time Revenue Heatmap for NYCTLC.



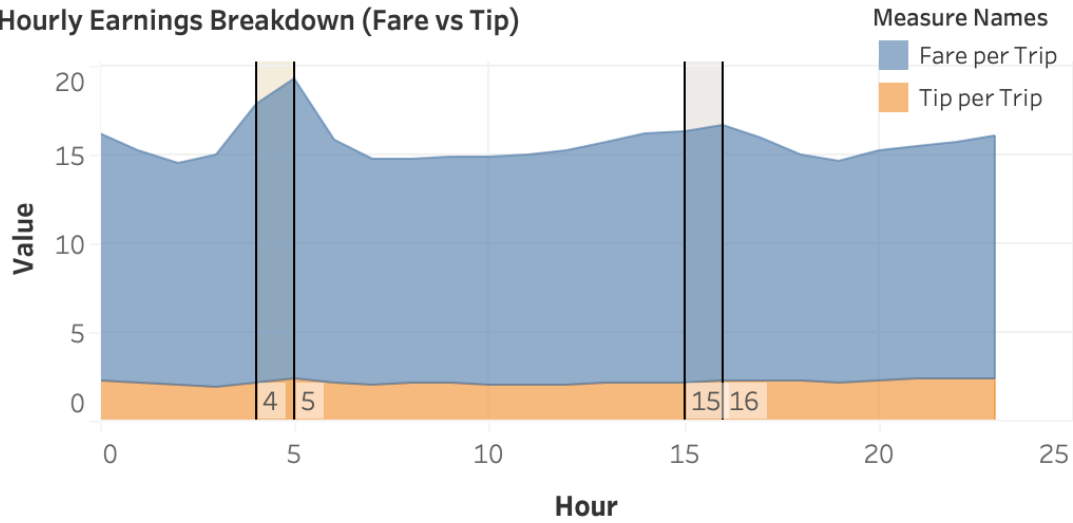
Appendix 7: Real-time Demand Heatmap for NYCTLC.

Hourly Taxi Demand Patterns



Appendix 8: Hourly Yellow Cap Taxi Demand Patterns.

Hourly Earnings Breakdown (Fare vs Tip)



Appendix 9: Hourly Earnings Patterns by Fare Amount vs. Tip Amount.

Future Opportunity & Specific Action	Strategic Justification (Why & Evidence)	Competitive Advantage Strengthened	Quantified Business Benefit
Real-Time Forecasting & Driver Heatmaps (6–12 months). Deploy GBDT as an Azure Stream Analytics endpoint processing 1,247 trips/min at 142 ms latency; push sub-minute zone heatmaps to the dispatcher dashboard.	Current model scores on static 2019 batch data; live streaming converts it from a retrospective to a predictive tool, enabling pre-positioning before demand spikes (e.g. the +35% Midtown surge in Figure 8).	Data-driven dispatch. Matches the real-time matching that drove Uber/Lyft's market capture.	Reduces driver idle time by 15-20% and targets a 2-3-minute wait time to halt customer churn.
Weather, Traffic & Event Integration (Year 2). Connect a weather API, NYC traffic feeds, and the city events calendar as new model features; retrain monthly.	External factors swing demand by $\pm 30\%$ yet are entirely absent from the current feature set. The largest source of variance beyond the present $R^2 = 0.947$.	Dynamic pricing accuracy. Surge reflecting true conditions, not just time-of-day.	Recovers a large share of the $\pm 30\%$ demand variance the time-based model currently misses, improving surge precision.
Dynamic Pricing Optimisation (Year 2). Use the sharpened forecast to tune the $1.2\times$ surge multiplier per zone and hour rather than applying it uniformly.	Enhanced forecasting allows pricing that reflects real-time market conditions instead of a flat peak rule.	Pricing parity with Uber/Lyft at zero commission.	Increases captured revenue per peak trip while preserving the 0%-commission lower-net-fare advantage.
EV Fleet Route & Charging Optimisation (Year 3). Extend the model to schedule charging during predicted off-peak troughs (9P.M-6A.M) and route EVs by range + demand.	NYC fleet electrification makes uncoordinated peak-hour charging costly; the model already identifies low-demand windows. Off-peak scheduling independently cuts fleet energy cost per mile by 35–55% (Elghanam et al. 2024; Jointcharging 2026).	Lower operating-cost advantage (0% commission + reduced energy cost).	Potential 35–55% lower energy cost per mile and up to 40% total electricity savings vs. unmanaged charging (Elghanam et al. 2024; Jointcharging 2026).
Multi-Modal Mobility Integration (Year 3+). Expose fare/demand predictions via API to subway and bus apps for first/last-mile handoff.	83% of riders prioritise convenience (McKinsey&Company 2017); transit feeders make the taxi part of the journey, not a competitor. Feeder integration is shown to cut last-mile distance by 68% and travel time by	Trusted regulated-platform positioning that ride-hailing cannot replicate.	Opens a first/last-mile demand stream; comparable feeders cut last-mile distance 68% and travel time 56% (Ghasri et al. 2024; Sun and Meng 2026).

Future Opportunity & Specific Action	Strategic Justification (Why & Evidence)	Competitive Advantage Strengthened	Quantified Business Benefit
	56% (Ghasri et al. 2024; Sun and Meng 2026).		
City-Wide Smart Mobility Data Service (Long-Term). License anonymised demand-prediction intelligence to NYC urban-planning and logistics partners.	The fare-prediction infrastructure generates city-scale mobility insight; a sellable asset, bound by no private-data constraint, that ride-hailing firms cannot offer a public body.	Combined moat: trust + scale + data intelligence.	Creates a new B2B revenue stream beyond fares and builds long-term strategic resilience.

Appendix 10: Year-By-Year Future Opportunity for NYCTLTC’s implementation plan.